UNIT 1 · 8 hours

Open-Loop and Closed-Loop Control Systems



Introduction and Mathematical Modelling of Physical Systems

Control Systems (22EEG53)
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Learning objectives

- Define a control system and its basic terms
- Distinguish open-loop from closed-loop systems with examples
- Derive the closed-loop transfer function
- Explain the effects of feedback on gain and sensitivity

Syllabus context

Introduction and mathematical modelling of physical systems: open loop and closed loop systems, examples, control system components. Transfer function of physical systems: mechanical systems (translational and rotational), electrical systems, analogous systems, block diagram reduction techniques, signal flow graphs.

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What is a control system?

Definition: an interconnection of components that makes a plant (process) behave in a desired way.

Input (reference) r: desired value of the output.

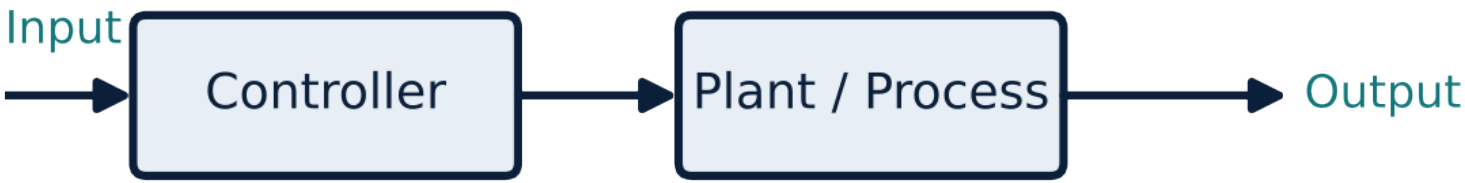
Output (controlled variable) c: the quantity being controlled.

Plant: the system to be controlled; controller: the component that generates the actuating signal.

Disturbances: unwanted inputs that tend to move the output away from the desired value.

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Open-loop system

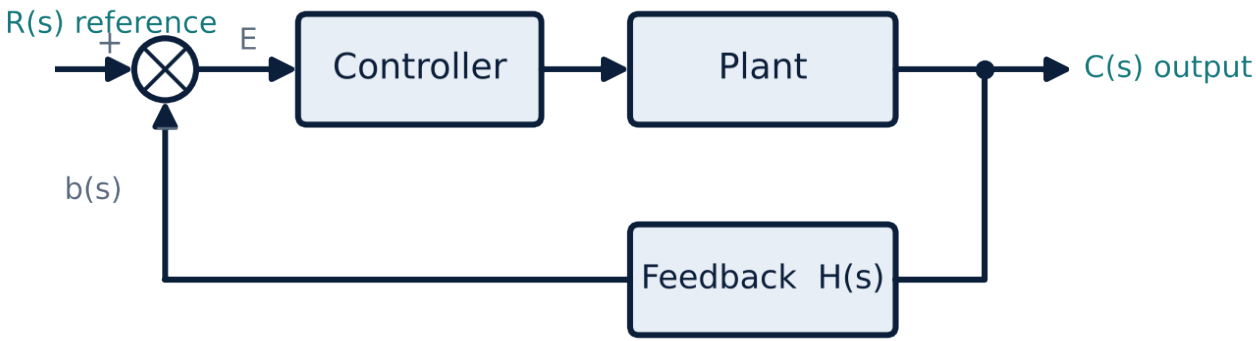


No feedback path

- Output has no effect on the control action.
- Accuracy depends on calibration of the controller.
- Cannot compensate for disturbances or parameter drift.

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Closed-loop (feedback) system



Negative feedback loop

- Output is measured and fed back.
- Error E = R − B = R − H·C drives the controller.
- Disturbances are corrected automatically.
- Examples: thermostat, cruise control, servo.

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Open loop versus closed loop

Feature	Open loop	Closed loop
Feedback	None	Present
Accuracy	Depends on calibration	High; self-correcting
Response to disturbance	No correction	Reduces effect
Stability	No problem	Can become unstable
Complexity and cost	Simple, cheap	More components, costly
Examples	Toaster, traffic signal	Thermostat, servo, cruise control

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Deriving the closed-loop transfer function

1

Forward path

$C(s) = G(s)\,E(s)$

2

Error

$E(s) = R(s) - B(s)$

3

Feedback

$B(s) = H(s)\,C(s)$

4

Eliminate E and B

$C = G(R - HC) \Rightarrow C(1 + GH) = G\,R$

$C(s)/R(s) = G(s) / [1 + G(s)H(s)]$ (negative feedback); $G/(1 - GH)$ for positive feedback

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Effect of feedback on sensitivity

Problem G = 100 and H = 0.1. Find the closed-loop gain; then find it if G falls by 10 % to 90.

- T = G/(1 + GH) = 100/(1 + 10) = 9.09
- With G = 90: T = 90/(1 + 9) = 9.00
- Change in T = (9.09 − 9.00)/9.09 ≈ 1 % for a 10 % change in G

Answer: Feedback reduces sensitivity: 10 % change in G → ≈ 1 % in T

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Effects of feedback

Gain

Overall gain reduced by 1/(1 + GH) for negative feedback.

Sensitivity and accuracy

Less sensitive to parameter variations and disturbances; steady-state error reduced.

Stability and bandwidth

Bandwidth increases, but too much feedback can cause oscillation or instability.

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Key takeaways

- Closed loop = feedback; open loop = no feedback
- T(s) = G/(1 + GH) for negative feedback
- Feedback reduces sensitivity to parameter changes and disturbances
- Feedback can cause instability: stability analysis is essential

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Practice questions

Q1. Differentiate open-loop and closed-loop systems with two examples each.

Q2. Derive C/R = G/(1 + GH).

Q3. G = 50, H = 0.2: find the closed-loop gain.

Q4. Why does feedback improve accuracy but risk instability?